Migrating Pipelines from ADF to Synapse

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1. Introduction

This document outlines the project undertaken to migrate existing Azure Data Factory (ADF) pipelines and Databricks notebooks to modern platforms: Azure Synapse and Microsoft Fabric, respectively. This modernization enhances scalability, maintainability, and integration within Azure's ecosystem.

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2. Problem Statement

The current system uses Azure Data Factory for data movement and transformation and Databricks for processing. The aim is to modernize the solution by migrating pipelines, datasets, and notebooks to Azure Synapse and Fabric, maintaining the same functionality while optimizing performance.

3. Objectives

3.1 Objective 1: ADF to Synapse Migration

- Develop ADF pipelines that:
    - Copy multiple tables from Azure SQL to ADLS Gen2.
    - Use Dataflows to clean the data.
- Migrate these pipelines, datasets, and linked services to Azure Synapse Studio in two different ways (manual + ARM Template or JSON export/import).
- Update connections post-migration and validate pipeline functionality.

3.2 Objective 2: Databricks to Fabric Migration

- Develop Databricks Notebooks.
- Migrate these notebooks to Microsoft Fabric.
- Update mount points/connections as needed.
- Validate each notebook post-migration.

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4. Tools & Technologies

- **Azure Data Factory** – Pipeline orchestration
- **Azure Synapse Analytics** – Unified analytics platform
- **Azure SQL Database** – Source data
- **Azure Data Lake Gen2** – Storage layer
- **Azure Key Vault** – Secrets management
- **Databricks** – Notebook development
- **Microsoft Fabric** – Notebook execution post-migration

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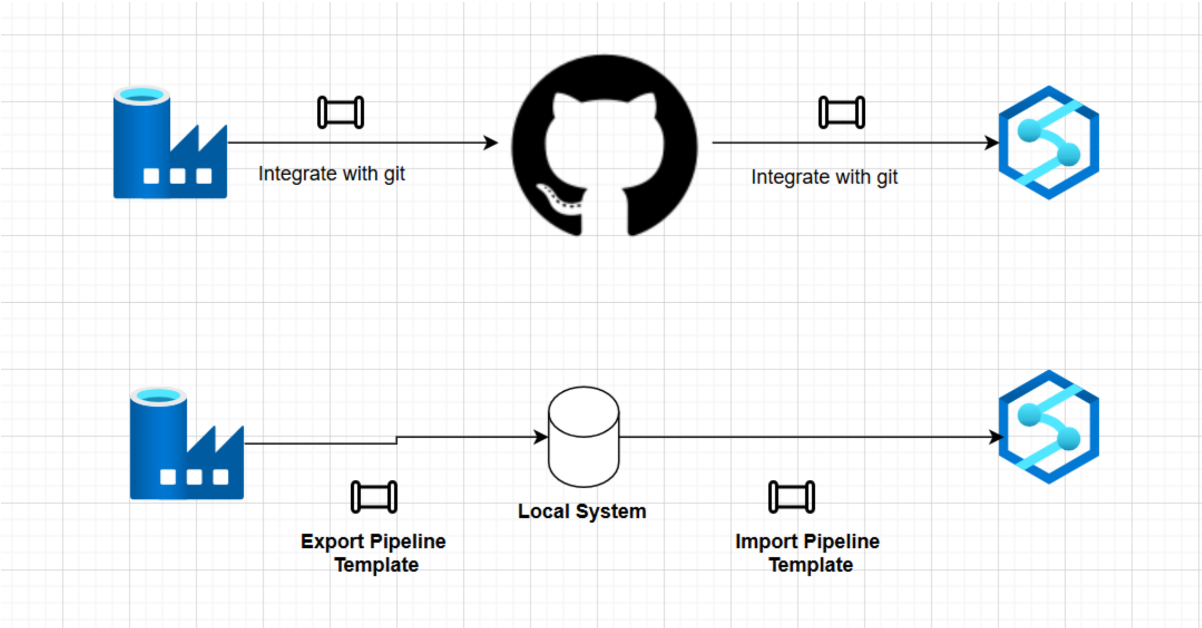
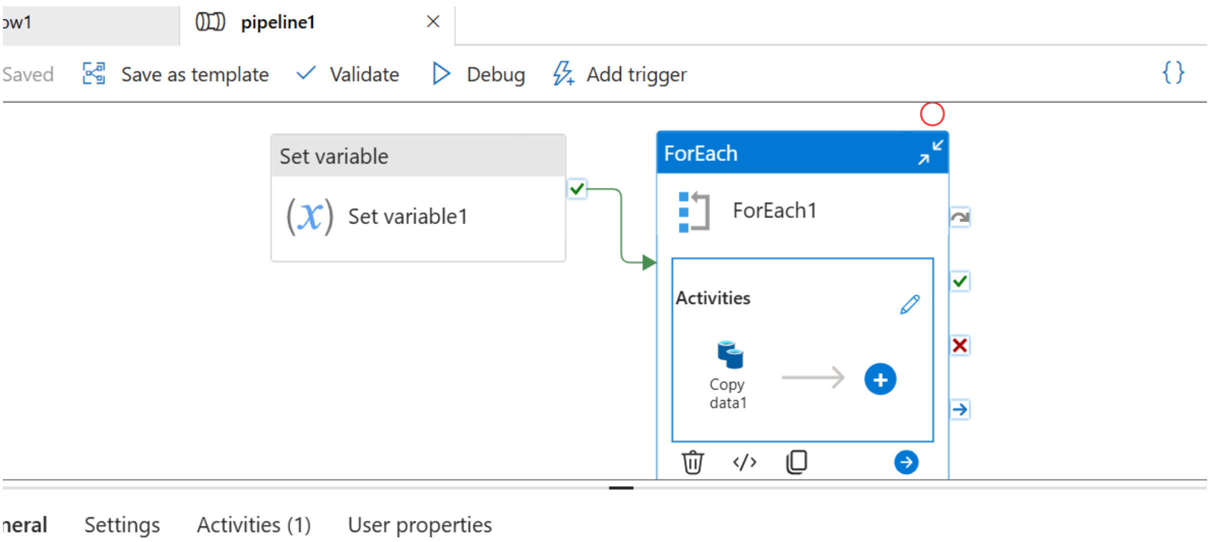
5. Architecture Diagram



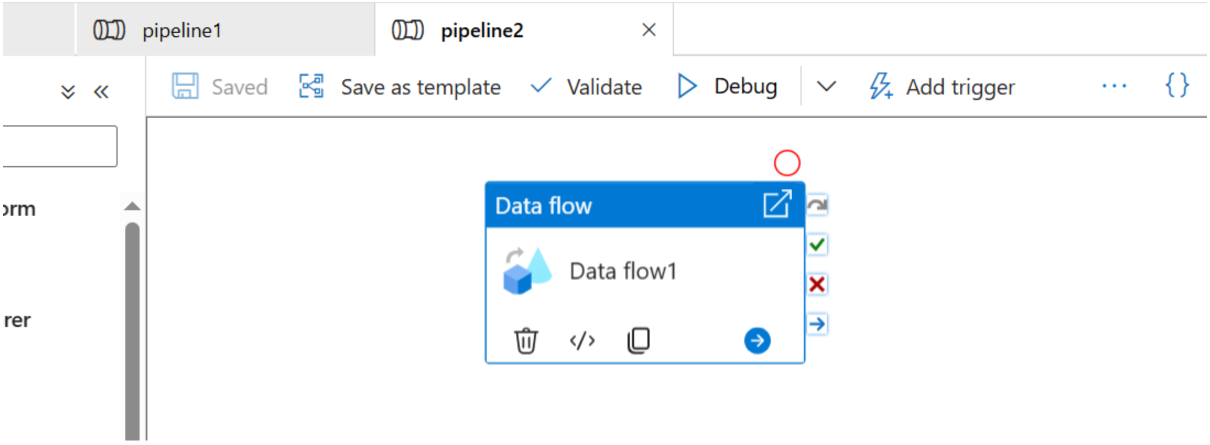
6. Implementation Details

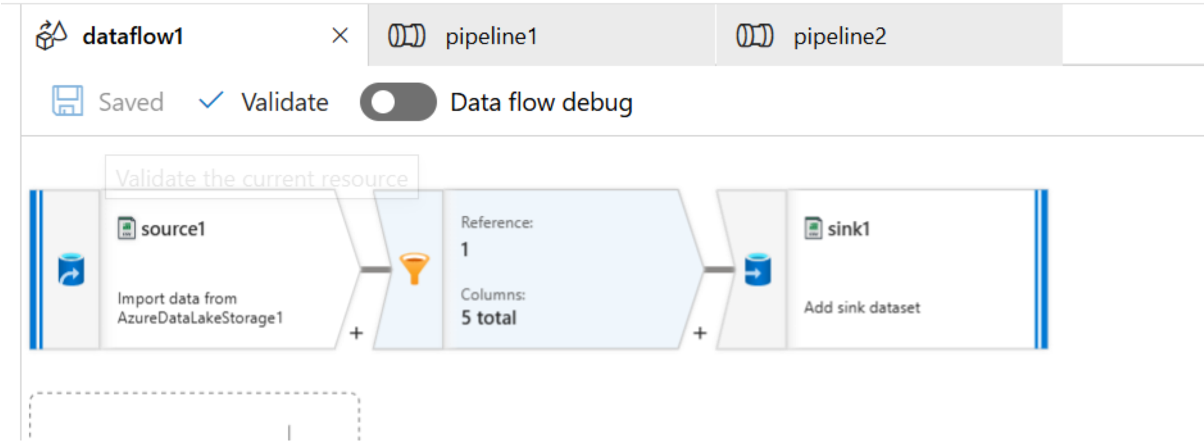
6.1 ADF Pipeline Development

- **Pipeline 1**: CopyMultipleTables using Lookup + ForEach + Copy Activity



- **Pipeline 2**: CleanDataWithDataflows using Mapping Dataflows

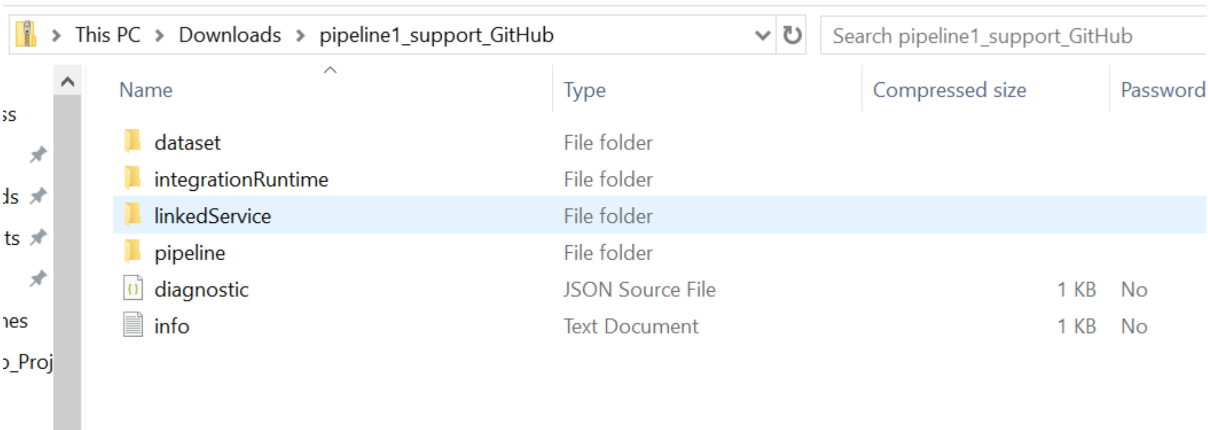




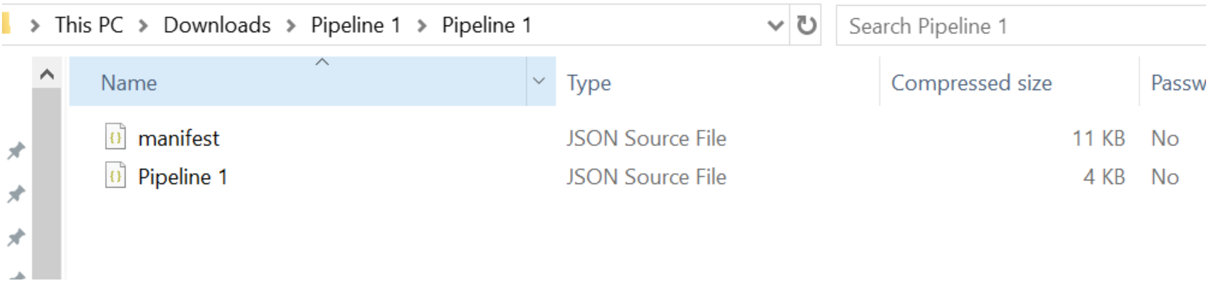
- **Linked Services**: Azure SQL, ADLS Gen2, Key Vault
- **Datasets**: Parameterized datasets for reuse

6.2 ADF to Synapse Migration Steps

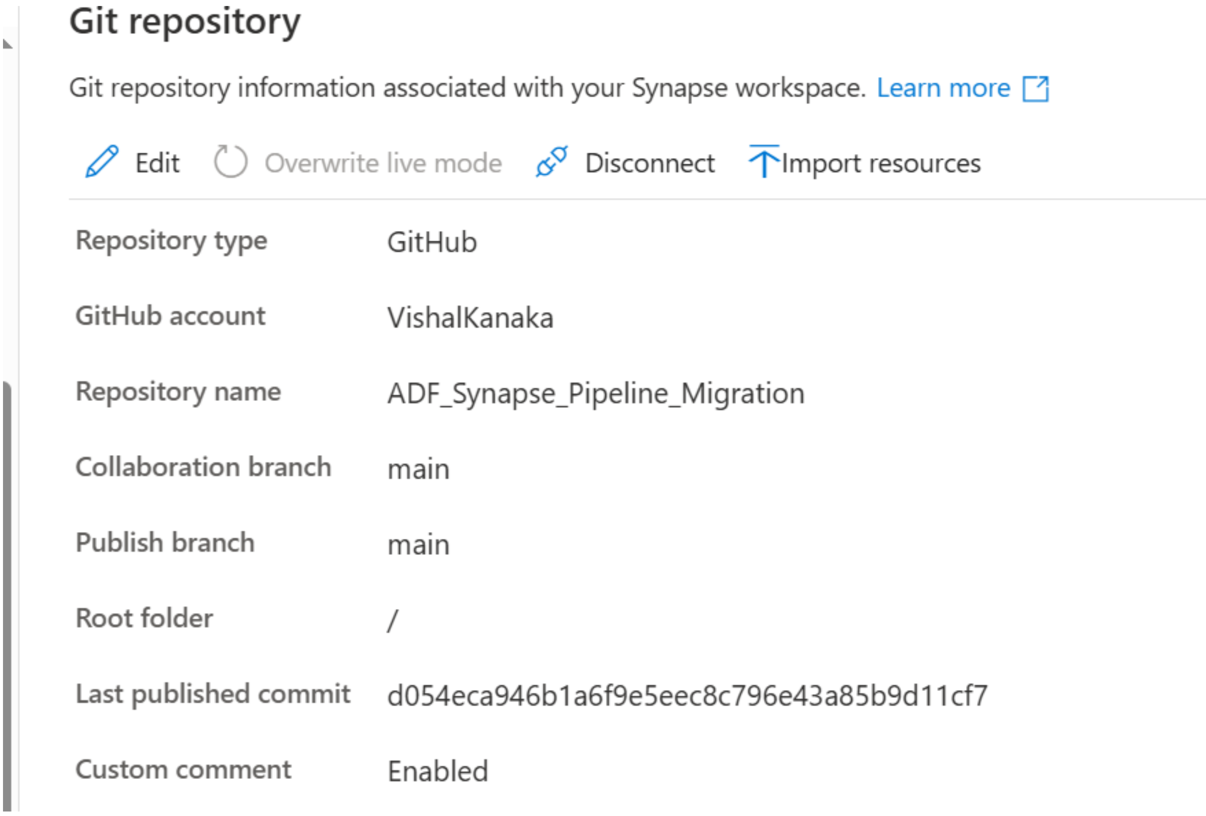
- **Method 1**: Manual recreation in Synapse Studio using JSON export/import



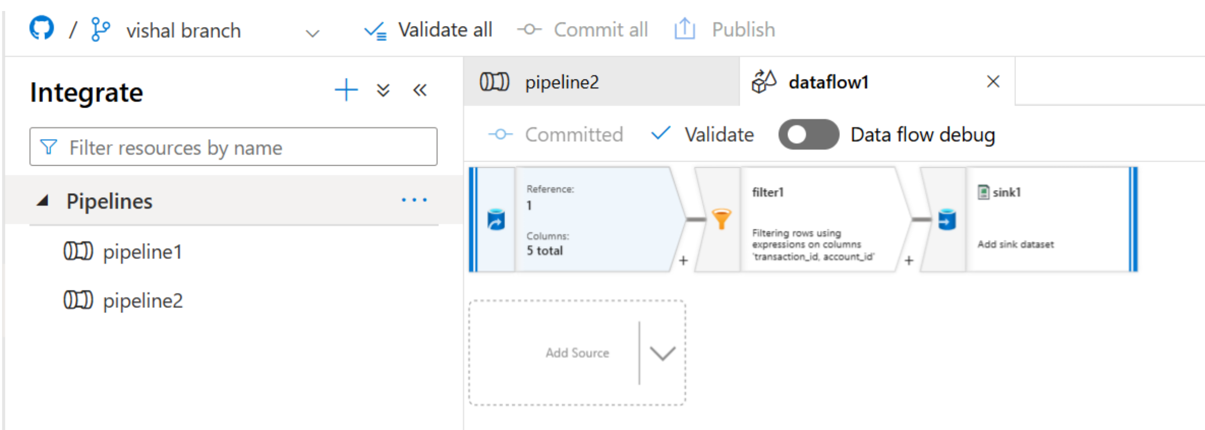
- **Method 2**: ARM Template deployment
- Export pipeline template



- **Method 3: Import from Github**



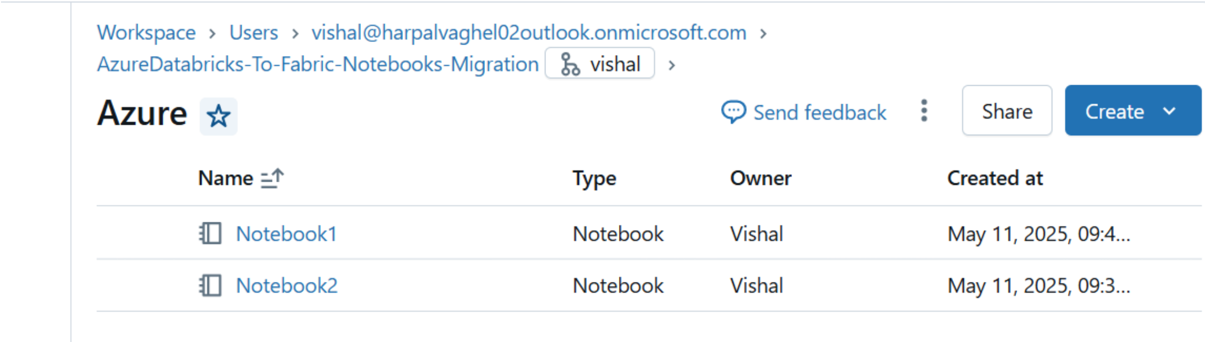
- Successfully imported in synapse

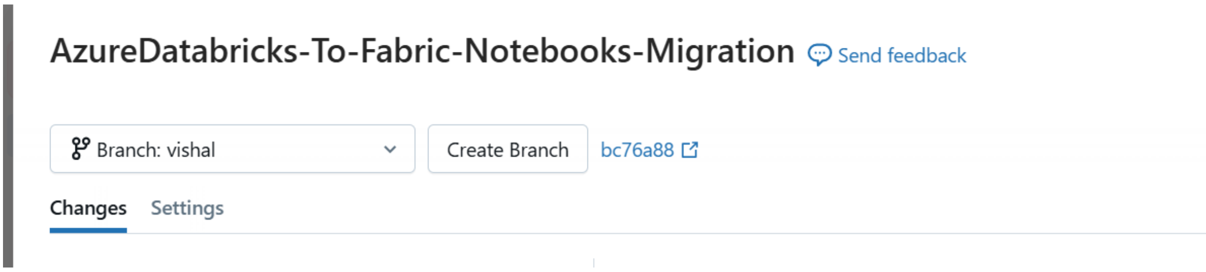


- **Post-Migration Steps**:
  - Replace linked services with Synapse equivalents
  - Update Key Vault references
  - Run integration tests

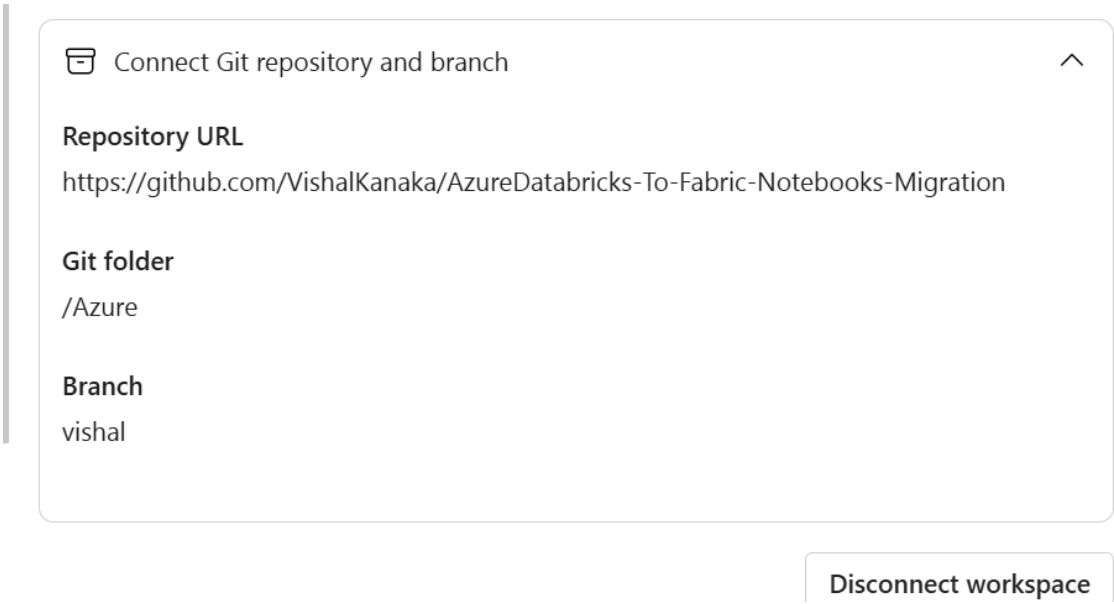
6.3 Databricks to Fabric Migration Steps

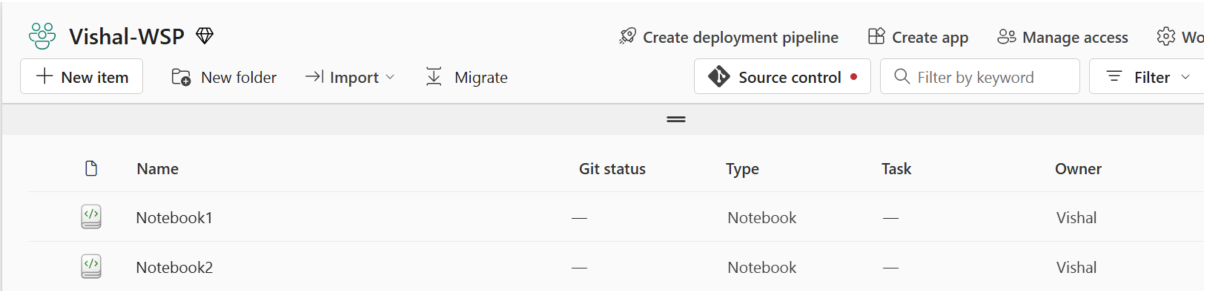
- Export notebooks from Databricks workspace (.dbc oripynb)



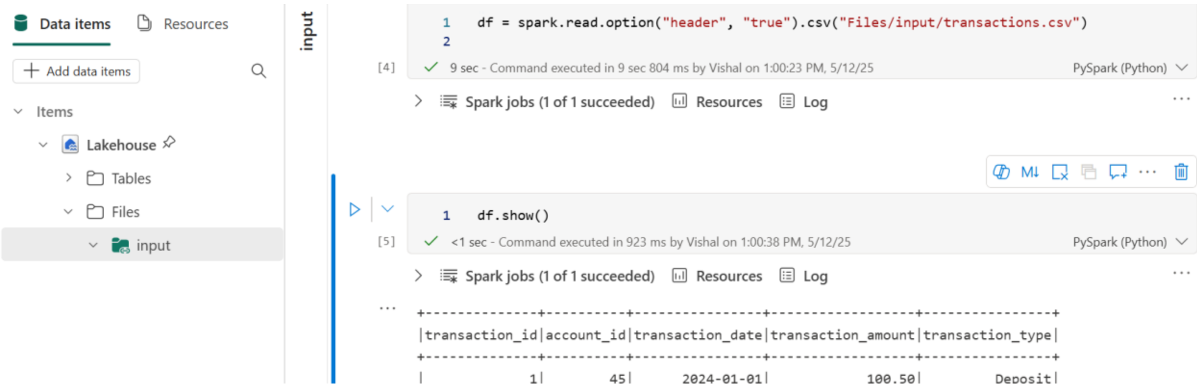


- Import into Microsoft Fabric from github



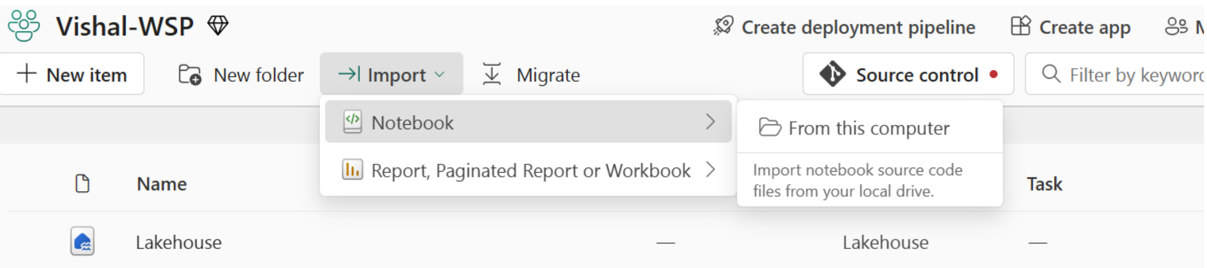


- Modify or remove dbutils.fs.mount() logic if unsupported



Method 2

- Manual import by uploading the notebooks



7. Deliverables

- ✅ Pipelines in ADF and Synapse
- ✅ Four Notebooks in Databricks and Fabric
- ✅ Architecture Diagram (.drawio file)
- ✅ Complete Documentation:
    - Data Ingestion Setup
    - Dataflow Logic
    - Notebook Code
- ✅ Uploaded to GitHub: [https://github.com/VishalKanaka/AzureDatabricks-To-Fabric-Notebooks-Migration.git]